Fiery DataXPress Specification

FieryPath and FITS framework for printer clients



Revision History

Version	Change Log	Date	Author
1.0	Initial Draft	11/30/2023	Aloke Bordia
1.1	Added jobName to the print message sent to the FITS client	04/12/2024	Aloke Bordia
1.2	Updated FITS Sample Client documentation	04/23/2024	Scott Crosby

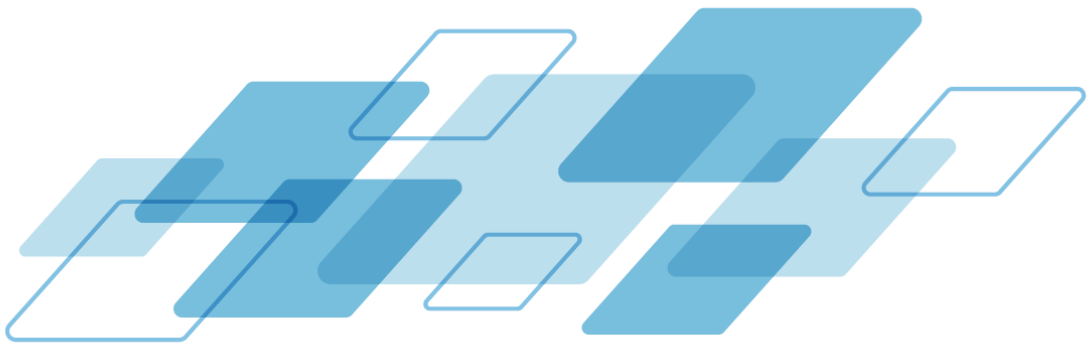




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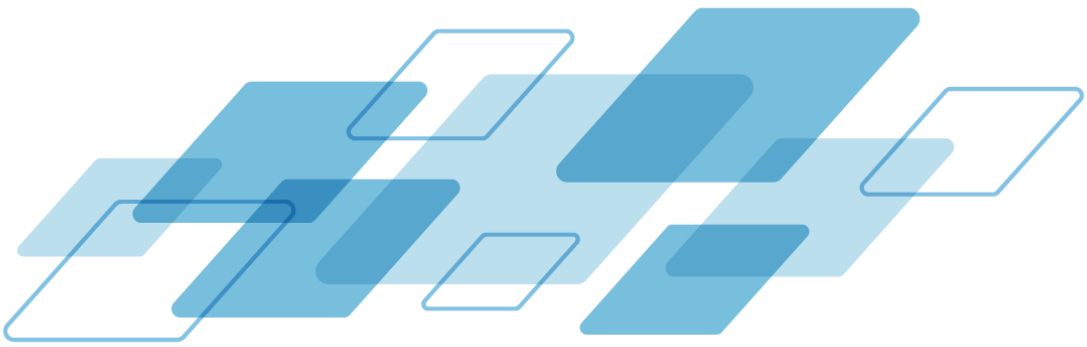
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1 Overview

This document is intended to serve as a reference for the printer to interface with Fiery DFE both for the Control Path (using FieryPath specification) and Data Path (using FieryPath Image Transfer Service specification). Fiery DataXPress (FDX) provides a framework that printer clients can leverage either to build upon the sample clients provided or use them as a reference to create their own clients. The goal of FDX is to demonstrate how the printer can create clients to print at speed.

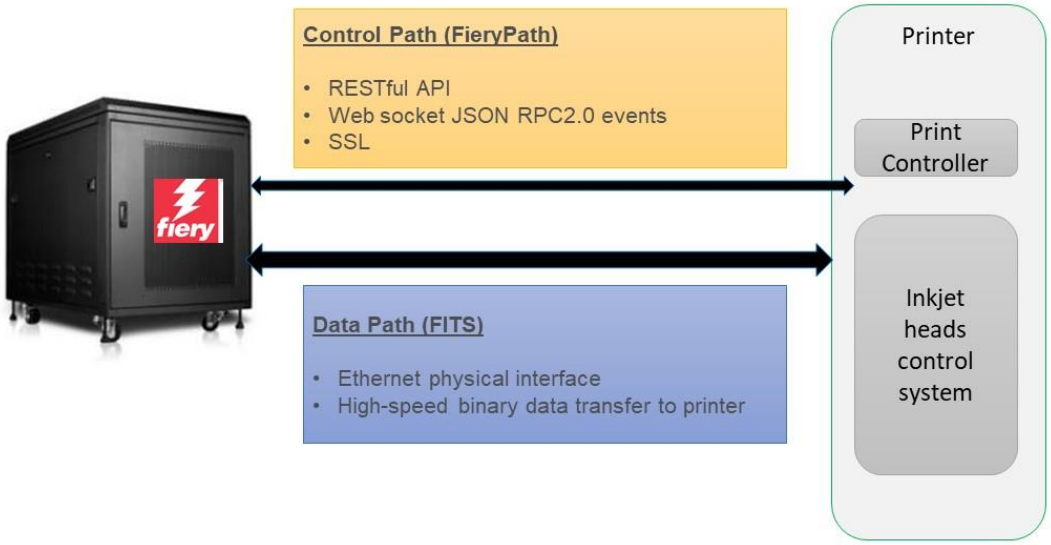
1.1 Control Path

The Control Path is the interface for handling all management/control functions. All client-initiated communication (i.e. actions/commands) is done over SSL through a RESTful API – with the DFE's Main Blade/Server acting as the HTTPS server and the printer acting as the client using the API defined in the FieryPath specification. All server-initiated communication (i.e. events) is done through a WebSocket interface – with the DFE acting as the WebSocket server publishing events to connected client. All events use the JSON RPC-2.0 protocol defined at http://www.jsonrpc.org/specification.

Refer to the FieryPath specification provided separately for more information.

1.2 Data Path

The Data Path is the interface for handling the transfer of image/ink data from the DFE to the printer. The Control Path, by receiving events and acknowledging them using the RESTful API, allows data to enter the Data Path. The FieryPath Image Transfer Service (FITS) specification provided separately gives more information about FITS and the protocol used for data transfer from the DFE to the printer.



2 Design

For Control Path, FDX will provide a sample client that will demonstrate the basic usage of FieryPath API. This will include making REST and websocket connection to Fiery server, send a few consumables and demonstrate the usage of print related events.

For Data Path, FDX will use a FITS-based client. This will include implementing the FITS protocol to connect and receive data from Fiery server. The sample FITS-Client will also demonstrate the use of buffering and communicate with the sample Control Path client using standard inter-process communication. In addition, the sample FITS-Client will provide several examples that will demonstrate “processing” the image data for final output (which could be writing to files or communicating with the electronics of your printer).

2.1 Print

The following sequence diagram illustrates the steps that the FieryPath and FITS-Module plugin should take to print an image.

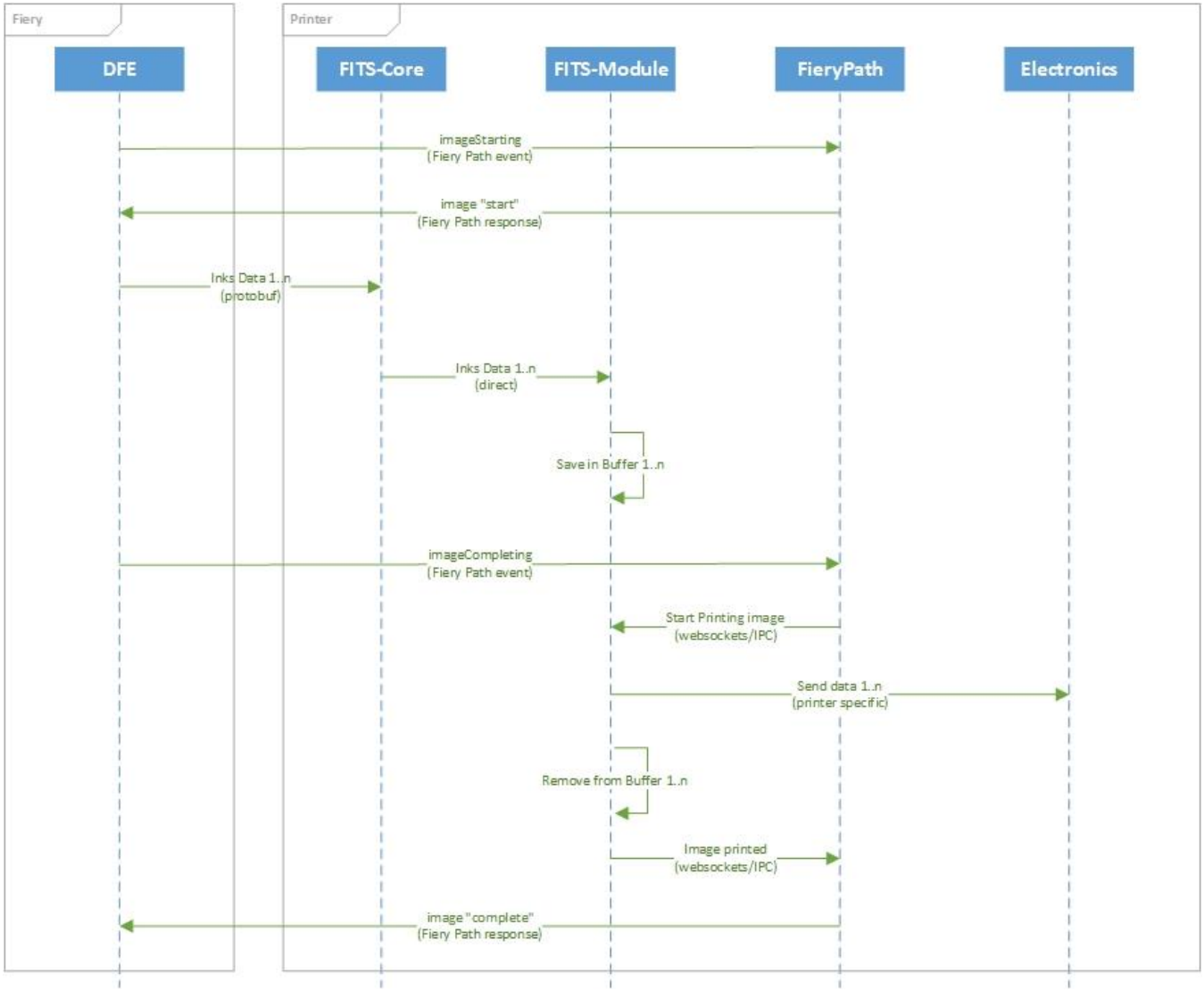


Figure 1: Print Sequence

The above sequence outlines the steps that the printer clients should take to print an image. The focus in this case is only on the data transfer and the subsequent steps required. For reference:

- DFE – represents the Fiery server that pushes image data using FITS protocol for 1..n inks
- FITS-Core – implements FITS protocol to receive data for 1..n inks and will load the FITS-Module plugin. The binary along with source code will be provided by Fiery. It will run on the printer software machine
- FITS-Module – receives data from FITS Core and sends data to print electronics when initiated. This module will be created by the printer team. It must use buffering to provide the most efficient way to print at speed
- FieryPath – implements the FieryPath API. Even if the printer does not require a full fledged FieryPath client, it should have a minimal client that helps to synchronize and initiate printing an image by sending messages to the FITS-Module
- Electronics – printer specific

Note that there can be multiple FITS-Core/FITS-Module clients based on the number of network interfaces. For the purposes of this document, a single FITS-Core/FITS-module is used to illustrate the design and usage.

As seen above, FITS-Core will receive the image data from Fiery server using FITS protocol. On receiving the data, FITS-Core will invoke a specific callback in FITS-Module. This will be done for each ink for which the data has been received. FITS-Module plugin will save the image data for each ink in its buffer and return immediately. The buffer can be heap memory, shared memory or the most efficient memory mechanism to transfer data to electronics when the image should be printed. It is important that no additional processing be done in this callback context other than saving the image data in the buffer. This will ensure the printer can print at speed without artificially throttling data transfer.

Once the data has been transferred to FITS-Module, “imageCompleting” event will be sent by Fiery to the FieryPath client. As has been noted throughout the FieryPath specification and Developer Guide, the printer should initiate printing of an image only once it receives the “imageCompleting” event. In the above case, the FieryPath client will send an internal event to the FITS-Module to initiate printing once it receives the “imageCompleting” event. For this communication, websocket is used in the samples provided but the printer can choose to implement any appropriate mechanism for this internal inter-process communication.

On receiving the internal message to print an image, FITS-Module should take the image data saved in the buffer for that image and send it to print electronics. This is specific to each printer. Once the image has been printed/stacked/ejected, FITS-Module should send a message to the FieryPath client indicating that the image has been printed.

On receiving the printed message from FITS-Module, FieryPath client should send an image “complete” response to the DFE for the “imageCompleting” event it had received. This will indicate to the DFE that the image has been successfully printed/stacked/ejected.

2.2 Cancel

The following sequence diagram illustrates the steps that the FieryPath and FITS-Module plugin should take to cancel a job.

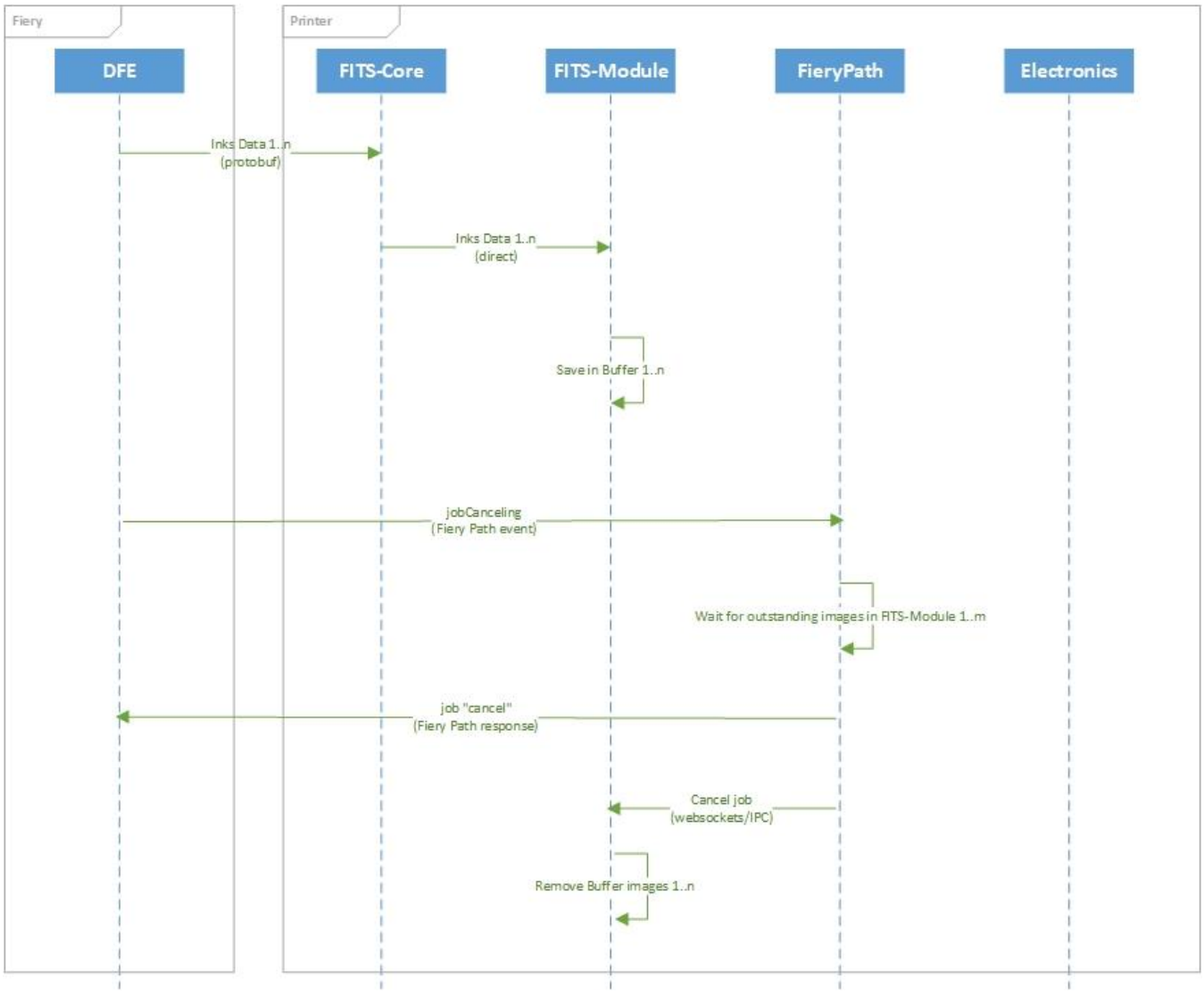


Figure 2: Cancel Sequence

When FieryPath client receives “jobCanceling” event, it should wait to print any outstanding images for which “print” message has been sent from the FieryPath client to the FITS-Module plugin. Once those images are printed/stacked/ejected, FieryPath client should respond with a job “cancel” response to the DFE.

Additionally, the FieryPath client should also send a message to the FITS-Module plugin indicating that the job has been canceled. This will allow the FITS-Module plugin to cleanup the image data in the buffer for that job. When an image is printed, the buffer will be cleaned up but, in the case when image data has been transferred to the FITS-Module plugin, but the printing was not initiated for it because the job was in the process of being canceled, there can be leftover image buffer data in the FITS-Module plugin for that job. Hence it is necessary for this message to be sent from the FieryPath client to the FITS-Module plugin so that it can do the necessary cleanup.

2.3 Internal Communication

As noted above, image data should be sent to the printer electronics only once the FieryPath client receives “imageCompleting” event. This guarantees that the data for that image has been transferred to the all FITS-Module buffers. To satisfy this requirement, FieryPath client and FITS-Module plugin need bidirectional communication.

It is completely up to the printer software to decide how they want to implement this communication. The samples provided as part of the FDX framework use websocket for this purpose. The following sequence diagram illustrates this communication.

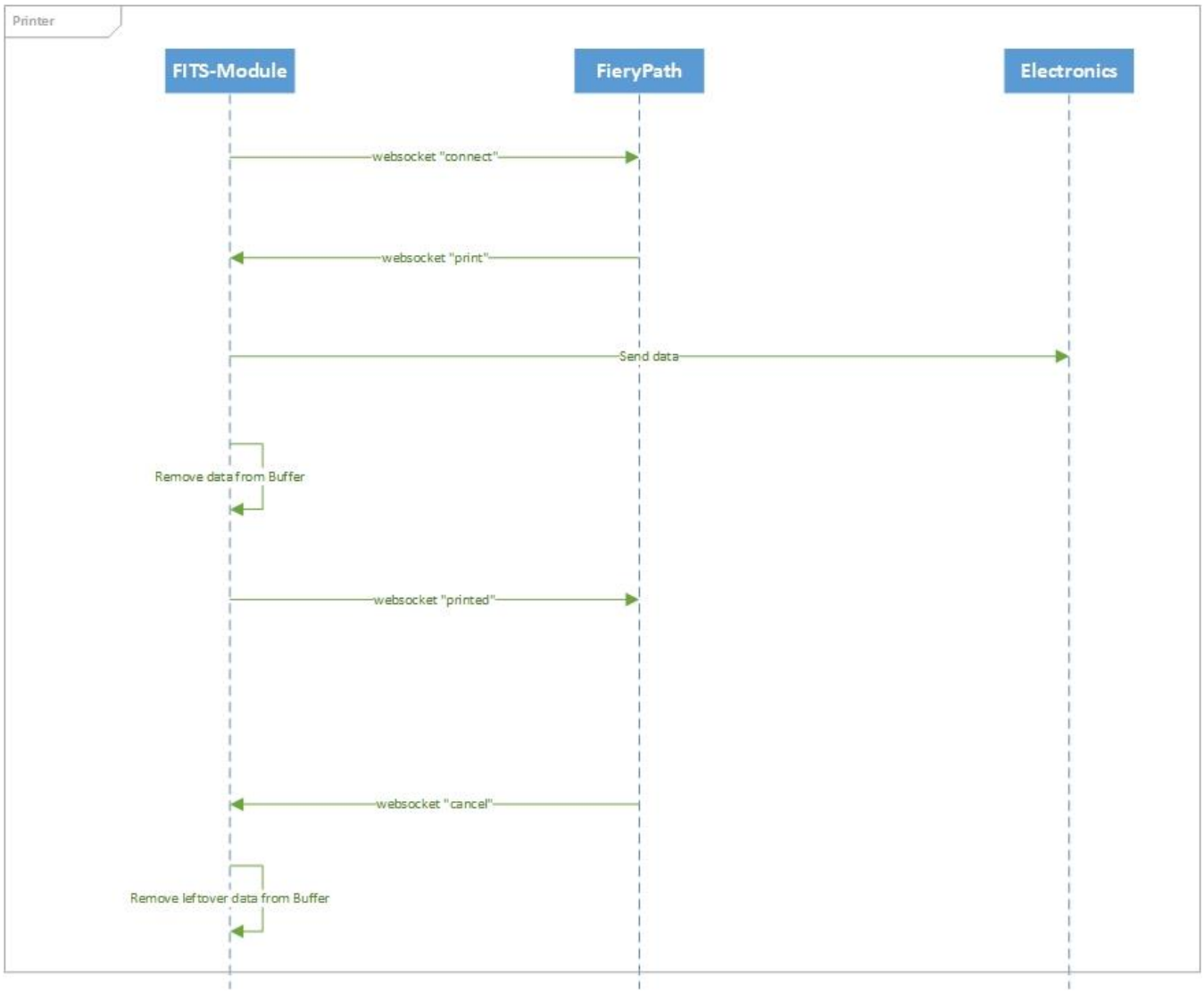


Figure 3: FieryPath to FITS-Module Plugin Sequence

For the samples provided with FDX, FieryPath client starts a websocket server on port 9085. This does mean that depending on where the clients are running, this port may have to be opened through the firewall. Once the FITS-Module plugin has information about the inks supported on the interface, it will send a websocket “connect” message to the FieryPath client. It is important to note that the FITS-Module plugin should initiate this websocket connection only once it has received the ink information from FITS-Core.

When FieryPath client receives “imageCompleting” event, it will send a websocket “print” message to the FITS-Module plugin. It is at this point that the FITS-Module plugin will take the data for that image saved in its buffer and send it to the printer electronics. Once the image has been printed/stacked/ejected, FITS-Module plugin will remove the data for that image from its buffer and send a websocket “printed” message to FieryPath client indicating that the image has been printed.

As noted above, FITS-Module plugin may also have leftover image data in its buffer when a job is canceled. For this reason, FieryPath client will send a websocket “cancel” message to FITS-Module plugin so that it can clean up its buffer. There is no response expected back for this message.

The samples use the following format for each of these messages. Printer clients can choose to use these or any other format they deem appropriate for this internal communication.

2.3.1 connect

Each FITS-Module plugin should send a “connect” message to FieryPath client once it has the information about the inks supported on that interface. There should be as many “connect” messages as the number of FITS-Module plugin clients. This is the format used by the sample client for this stringified message:

{ type: “connect”, inks: [“cyan”, “magenta”, “yellow”, “black”] }

2.3.2 print

When FieryPath client receives “imageCompleting” event, it will notify all the FITS-Module plugins that are connected to “print” that image. This is the format used by the sample client for this stringified message. The reason “inks” is sent with this message is so that the FITS-Module plugin client knows which inks are part of this image and should be printed.

{ type: “print”, jobId: jobId1, imageId: imageId1, jobName: jobName1, inks: [“cyan”, “magenta”, “yellow”, “black”] }

2.3.3 printed

Once the FITS-Module plugin has printed an image, it should send a “printed” message back to the FieryPath client. All FITS-Module plugins should send this message. This is the format used by the sample client for this stringified message:

{ type: “printed”, jobId: jobId1, imageId: imageId1 }

2.3.4 cancel

When a job is canceled, FieryPath client will send a “cancel” message to all FITS-Module plugin clients to cleanup any buffers it may have that are used for images in the job. This is the case when image data may have been transferred to FITS-Module plugins, but the printing of the image was not initiated because the job was canceled. This is the format used by the sample client for this stringified message:

{ type: “cancel”, jobId: jobId1 }

3 Package

In addition to the FieryPath and FITS specifications, the following will be provided as part of the FDX workflow.

3.1 FieryPath sample client

FieryPath client can be written in any language that supports creating a RESTful client. The sample provided with FDX is written in TypeScript. It can be used as a starting point to create a full-fledged FieryPath client or as a reference to create a client in another language.

For more details about the client implementation, refer to the readme file provided with the sample client.

3.2 FITS sample client

The sample FITS-Client is a C++ implementation using the FITS protocol to connect and receive data from Fiery server. The sample FITS-Client will also demonstrate the use of buffering and communicate with the sample Control Path client using standard inter-process communication. In addition, the sample FITS-Client will provide several examples that will demonstrate “processing” the image data for final output (which could be writing to files or communicating with the electronics of your printer). It can be used as-is, as the framework to implement your printer-specific protocol necessary to transfer image data to your electronics, or as the starting point for your own design.

For more details, refer to the **FieryPath_DataXPress_Sample_FITS-Client** document.



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